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Digitally Controlled Mixed-Signal Power Supply

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1. Synopsis

This report documents the design of an adjustable linear bench power supply built around an Arduino Nano Every, a TLC5618A dual-channel DAC, and a 320×240 ILI9341 SPI TFT display. The supply uses a hybrid digital/analog control architecture: the user sets a target voltage and current limit via two rotary encoders, the Arduino converts that target into an analog reference voltage through the DAC, and a dedicated analog op-amp control loop regulates the actual output using series-pass transistors — the microcontroller never sits directly inside the high-current regulation path.

The design integrates rectification and bulk filtering, dual isolated buck-regulated auxiliary rails for clean control-electronics power, a compensated analog voltage control loop, a differential-amplifier current sense stage, and a digit-addressable rotary-encoder user interface with live TFT readout of set and measured voltage/current.

2. Introduction

Bench power supplies are a foundational tool for hardware development and debugging, and building one from discrete parts is a practical way to work through analog regulation, feedback control, mixed-signal interfacing, and embedded firmware design in a single project.

This design deliberately separates two concerns that are often conflated in simpler DIY supplies:

- **Digital setpoint generation** — the Arduino and DAC only decide what voltage/current is wanted; they never carry the load current or sit inside the fast feedback path.
- **Analog regulation** — a dedicated op-amp control loop, compensated for stability, does the actual real-time work of holding the output at the requested value, using series-pass transistors.

This separation means a firmware bug, a slow SPI transaction, or a display refresh can never destabilize the actual output regulation — a property worth having in any tool you plan to trust your other projects to.

3. System Overview

3.1 Block Diagram

Figure 1 shows the overall signal flow: mains AC is rectified into a high-current DC bus, two buck converters derive clean low-voltage rails for the digital and analog control electronics, the Arduino/DAC pair generates the setpoint, and the analog error amplifier closes the loop around the series-pass output stage. Current is sensed independently via a shunt and differential amplifier, and both voltage and current readings are displayed on the TFT.

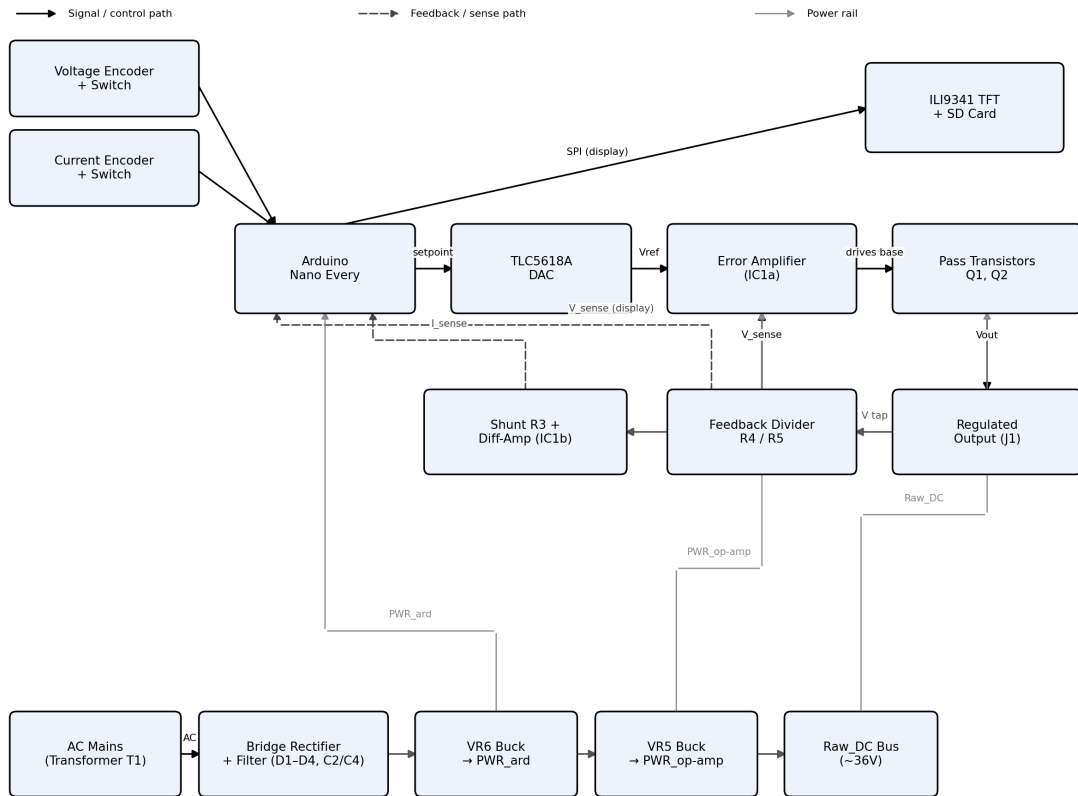


Figure 1: System block diagram.

3.2 Schematic Diagram

Figure 2 reproduces the full circuit schematic as designed in the PCB CAD tool, covering the AC input, dual buck auxiliary rails, DAC and Arduino interfacing, analog control loop, current-sense amplifier, and output stage.

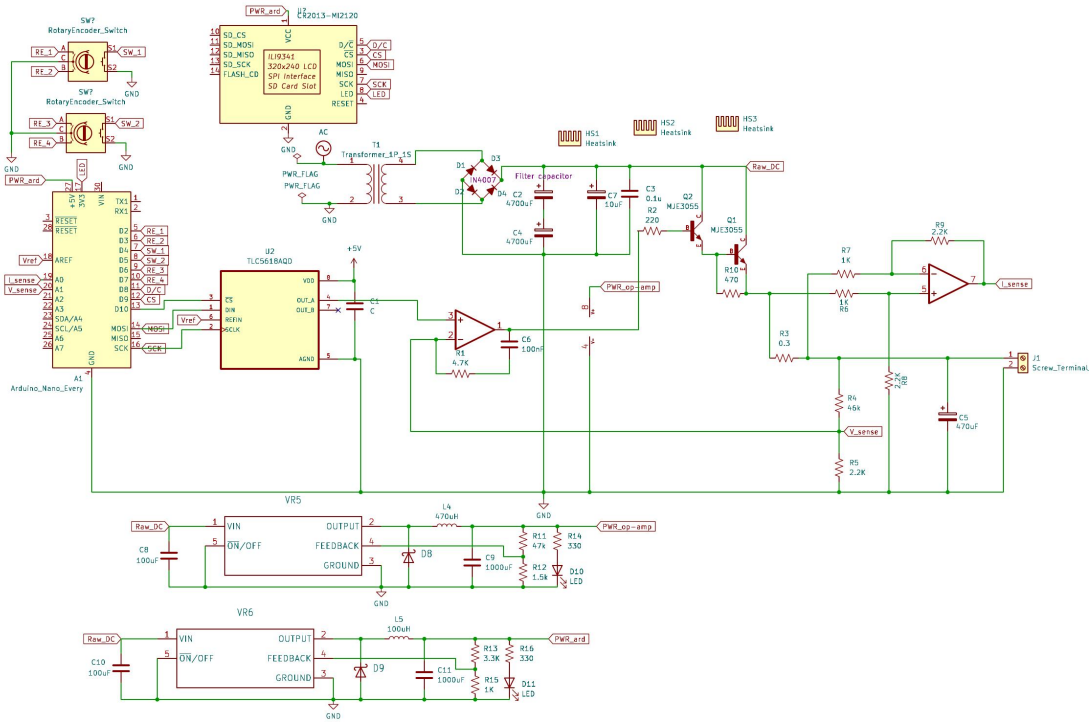


Figure 2: Full circuit schematic.

3.3 Pin Assignment Table

Table 1 summarizes the Arduino Nano Every pin assignments used in this design, confirmed against the schematic.

Pin	Signal	Connected To	Function
D2	RE_1	Voltage encoder A	Interrupt input, quadrature phase A
D3	RE_2	Voltage encoder B	Quadrature phase B
D4	SW_1	Voltage encoder switch	Selects digit place being edited
D5	SW_2	Current encoder switch	Selects digit place being edited
D6	RE_3	Current encoder A	Interrupt input, quadrature phase A
D7	RE_4	Current encoder B	Quadrature phase B
D8	D/C	ILI9341 TFT	Data/Command select

Pin	Signal	Connected To	Function
D9	CS	ILI9341 TFT	Chip select
D10	CS (DAC)	TLC5618A DAC	Chip select
D11	MOSI (hw SPI)	DAC + TFT (shared bus)	SPI data out
D12	MISO (hw SPI)	DAC + TFT (shared bus)	SPI data in (unused by both)
D13	SCK (hw SPI)	DAC + TFT (shared bus)	SPI clock
A0	I_sense	Current-sense diff-amp output	Analog input, output current readback
A1	V_sense	Voltage feedback divider (R4/R5)	Analog input, output voltage readback
AREF	Vref	Shared DAC REFIN / internal ADC ref	1.090 V reference, shared by DAC and ADC
5V	VCC	All digital peripherals	Supply voltage (PWR_ard)
GND	GND	Entire circuit	Common ground

Table 1: Arduino Nano Every pin assignment.

Note: the two rotary encoder modules and the ILI9341 TFT/SD card module are off-board panel components, wired in via headers rather than populated as footprints on the main PCB — confirmed against the interactive BOM, which lists no footprints for these parts.

4. Circuit Design & Working Principle

4.1 AC Input and Rectification

An off-board mains transformer (T1) connects to the board via a 2-pin screw terminal footprint. Its secondary AC feeds a full-wave bridge rectifier (D1–D4, 1N4007), producing a pulsating DC bus of approximately 36 V peak.

Bulk smoothing is provided by C2 and C4 (4700 μ F, 25 V each), deliberately wired in series. Since the rectifier output (~ 36 V) exceeds a single capacitor’s 25 V rating, the series connection doubles the effective voltage rating to 50 V, at the cost of halving the effective capacitance to approximately 2350 μ F. This produces the unregulated Raw_DC rail that feeds everything downstream.

4.2 Auxiliary Regulated Rails

Two independent LM2575-class buck regulators step Raw_DC down to two separate, clean low-voltage rails — physically isolating the digital control electronics’ switching noise from the sensitive analog control loop:

- **VR5** $\rightarrow$ **PWR_op-amp** — powers the dual op-amp (IC1). Feedback set by R11/R12, with L4 as the buck inductor, D8 as the freewheeling diode, and C8/C9 as input/output filtering.
- **VR6** $\rightarrow$ **PWR_ard** — powers the Arduino Nano Every. Feedback set by R13/R15, with L5 as the buck inductor, D9 as the freewheeling diode, and C10/C11 as input/output filtering.

Each rail has its own power-good indicator LED (D10 with R14 for VR5; D11 with R16 for VR6).

4.3 Digital Setpoint Generation

The Arduino Nano Every reads the two rotary encoders (with integrated push switches) and computes a target voltage and current limit. These setpoints are sent over SPI

to the TLC5618A dual-channel DAC (U2); channel A's output (OUT_A), filtered by C1, becomes the analog reference voltage V_{ref} that the control loop regulates toward. Channel B (OUT_B) is left unconnected in the current design, reserved for future use.

4.4 Analog Voltage Control Loop

The actual voltage regulation is performed entirely in analog hardware by one half of the dual op-amp package (IC1), configured as an error amplifier:

- Non-inverting input (+): V_{ref} , from the DAC.
- Inverting input (−): V_{sense} , from the R4/R5 output feedback divider.
- Feedback network: R1 (4.7 k Ω) in series with C6 (0.1 μ F), forming a Type II compensator.
- Output: drives the base of the parallel series-pass transistors, Q1 and Q2 (MJE3055), both mounted on a shared heatsink (HS1).

The Type II compensator is essential for loop stability: C6 alone would act as a pure integrator (a pole at the origin), giving very high DC gain for accurate regulation but pushing phase toward -90° across the whole spectrum. R1 in series with C6 introduces a compensating zero, restoring phase margin near the loop's crossover frequency so the supply regulates accurately without oscillating.

Base-region resistors R2 (220 Ω) and R10 (470 Ω) bias the pass transistors; R10 additionally provides a fast discharge path for stored base charge, ensuring clean transistor turn-off and bleeding thermal leakage current that could otherwise cause voltage drift.

4.5 Current Sensing

Output current is measured independently of the voltage control loop, using the other half of the same dual op-amp package (IC1) as a classic four-resistor differential amplifier around a 0.3 Ω shunt resistor (R3):

- R6 (1 k Ω) and R7 (1 k Ω): matched input resistors from either side of the shunt.
- R8 (2.2 k Ω) and R9 (2.2 k Ω): matched feedback/reference resistors.
- Differential gain = $R9/R7 = 2.2\times$, referenced to the output rail rather than system ground.

The amplified signal (I_{sense}) is read directly by the Arduino's ADC for display and any software-based current limiting; it does not feed into the hardware voltage loop, so there is currently no hardware constant-current (CC) mode — only a software-implemented foldback approach, described in Section 6.4.

4.6 Output Stage

A 470 μF output capacitor (C5) absorbs load transients, keeping the output stable during sudden current demand changes. The regulated output is delivered to the load through a 2-pin screw terminal (J1).

5. Quantitative Design Analysis

This section works through the actual voltage and current relationships at each stage, using the calibration constants confirmed in the firmware ($\text{DAC_VREF} = \text{ADC_VREF} = 1.090 \text{ V}$, shared via a common reference net; $\text{V_OUT_GAIN} = 22.886$; $\text{I_SENSE_GAIN} = 0.7398 \text{ V/A}$).

5.1 DAC Output Range

$$V_{dac} = 2 \times \text{DAC_VREF} \times \frac{\text{code}}{4096} \quad (1)$$

At full-scale ($\text{code} = 4095$): $V_{dac,max} = 2 \times 1.090 \times (4095/4096) \approx 2.179 \text{ V}$. This is the maximum Vref the control loop can be asked to track.

5.2 Voltage Feedback Divider

$$V_{sense} = V_{out} \times \frac{R5}{R4 + R5} = V_{out} \times \frac{2.2}{48.2} \approx V_{out} \times 0.04564 \quad (2)$$

Since the op-amp forces $V_{sense} = V_{ref}$ at DC equilibrium, $V_{out} = V_{ref}/0.04564 \approx V_{ref} \times 21.91$. This resistor-derived gain ($21.91\times$) closely matches the firmware's independently calibrated V_OUT_GAIN ($22.886\times$), a good cross-check that the divider values are doing what the design intends.

Combining the two: the DAC's full-scale output could theoretically command up to $\sim 49.9 \text{ V}$, but this is never reached in practice — the firmware clamps the setpoint to 30 V , and Raw_DC itself ($\sim 36 \text{ V}$) limits the realistic ceiling to a few volts below that, accounting for pass-transistor dropout.

5.3 Buck Regulator Output Voltages

Per the regulator's datasheet, the adjustable buck output follows:

$$V_{out} = V_{ref} \times \left(1 + \frac{R2}{R1}\right), \quad V_{ref} = 1.23 \text{ V} \quad (3)$$

where R2 is the resistor from FEEDBACK to Vout, and R1 is the resistor from FEEDBACK to GND. For VR6 (R13 = 3.3 kΩ as R2, R15 = 1 kΩ as R1): $V_{out} = 1.23 \times (1 + 3300/1000) = 5.29 \text{ V}$ — a clean, sensible value for directly powering the Arduino’s 5 V pin.

For VR5, R2 = 110 kΩ and R1 = 4.7 kΩ give $V_{out} = 1.23 \times (1 + 110000/4700) \approx 30.02 \text{ V}$ — confirming the corrected feedback network achieves the intended 30 V rail well within Raw_DC’s 36 V headroom.

5.4 Current-Sense Path

$$V_{shunt} = I_{out} \times R3 = I_{out} \times 0.3; \quad I_{sense,ideal} = V_{shunt} \times \frac{R9}{R7} = I_{out} \times 0.66 \text{ V/A} \quad (4)$$

The resistor-predicted gain (0.66 V/A) is reasonably close to the firmware’s empirically calibrated ISENSE_GAIN (0.7398 V/A); the firmware’s measured value should be treated as authoritative for real current calculations, since it accounts for real-world component tolerances that the ideal resistor math does not.

5.5 ADC Readback

Because the firmware calls `analogReference(INTERNAL)`, the ADC’s full-scale reference is 1.090 V (the same physical reference feeding the DAC’s REFIN), not the board’s 5 V supply rail. A raw 10-bit ADC code therefore converts as:

$$V_{out,measured} = \left(\frac{\text{rawADC}}{1023} \times 1.090 \right) \times 22.886 \quad (5)$$

$$I_{out,measured} = \left(\frac{\text{rawADC}}{1023} \times 1.090 \right) / 0.7398 \quad (6)$$

6. Firmware / Program Logic

The firmware runs a fully non-blocking main loop — no `delay()` calls — so encoder input remains responsive while sensor reads and display updates happen on their own periodic schedules.

6.1 Encoder Input Handling

Both rotary encoders are decoded via hardware interrupts on their A-phase pins, using a 2-bit quadrature state-transition check to register clean up/down steps and reject contact bounce. Each encoder’s push switch cycles a “digit cursor” (thousandths → hundredths → tenths → ones → tens), and turning the knob adjusts the setpoint by that digit’s place value — letting the same control provide both fine and coarse adjustment.

6.2 DAC Control

The DAC is driven over hardware SPI using the confirmed 16-bit control-word format: the top nibble selects DAC A or DAC B and sets fast/normal power mode (0xC000 for DAC A, 0x4000 for DAC B), and the remaining 12 bits carry the output code. A dedicated `voltageToCode()` function inverts the DAC transfer function to compute the code needed for a requested output voltage.

6.3 Display Rendering

The TFT is driven via the `LCDWIKI_SPI` library over the same hardware SPI bus as the DAC, each with its own independent chip-select line (D9 for the TFT, D10 for the DAC) confirmed against the schematic and the physical board. The display shows Set V, Out V, Set I, and Out I, with a small cursor bar indicating which digit each encoder is currently editing, and a CV/CC status indicator.

6.4 Software Current Limiting

Since no hardware current-control loop exists on this board, the firmware implements a software foldback: if measured current exceeds the set limit, the firmware walks the DAC's voltage command down in small steps until current settles back within the limit, restoring the original setpoint once the overcurrent condition clears. This is deliberately conservative — it protects against sustained overcurrent but is far slower than a true analog CC loop and should not be relied upon for short-circuit protection.

7. User Interface & Operation

7.1 Voltage / Current Setpoint Adjustment

Two rotary encoders — one for voltage, one for current limit — let the user dial in a target value digit-by-digit, with the associated push switch selecting which digit is currently being edited.

7.2 Display Readout

The TFT continuously shows both the requested (Set) and actual (Out) voltage and current, refreshed on a fixed interval independent of encoder input, so the display never becomes a bottleneck for control responsiveness.

7.3 CV/CC Indication

A status indicator on the display shows CV (constant voltage) during normal operation, switching to CC (constant current) whenever the software current-limit foldback described in Section 6.4 is actively reducing the output to respect the current limit.

8. Bill of Materials

Table 2 lists every component used in the design, cross-referenced against the project's interactive BOM export and the schematic.

#	Component	Reference(s)	Specification	Qty
1	Transformer	T1	Transformer_1P_1S (off-board)	1
2	Bridge rectifier diodes	D1, D2, D3, D4	1N4007	4
3	Bulk filter capacitors	C2, C4	4700 μ F, 25 V (series-connected)	2
4	Buck regulator IC	VR5, VR6	LM2575-class adjustable buck	2
5	Buck inductors	L4, L5	470 μ H, 100 μ H	2
6	Buck flyback diodes	D8, D9	Generic diode (DO-41)	2
7	Buck filter capacitors	C8, C9, C10, C11	100 μ F / 1000 μ F	4
8	Buck feedback resistors	R11, R12, R13, R15	110K Ω 4.7K Ω	4
9	Status LEDs	D10, D11	5 mm LED	2
10	LED series resistors	R14, R16	330 Ω	2
11	Dual op-amp	IC1	8-pin DIP dual op-amp	1
12	Voltage feedback divider	R4, R5	46 k Ω , 2.2 k Ω	2
13	Loop compensation	R1, C6	4.7 k Ω , 0.1 μ F	2
14	Series-pass transistors	Q1, Q2	MJE3055 (NPN, TO-220)	2
15	Base resistors	R2, R10	220 Ω , 470 Ω	2
16	Heatsinks	HS1, HS2, HS3	TO-220 / TO-247 class heatsink	3
17	Current shunt	R3	0.3 Ω	1
18	Diff-amp resistors	R6, R7, R8, R9	1 k Ω , 1 k Ω , 2.2 k Ω , 2.2 k Ω	4
19	Output capacitor	C5	470 μ F	1
20	Output terminal	J1	2-pin screw terminal	1
21	DAC output filter	C1	Ceramic disc capacitor	1
22	DAC breakout	U2	TLC5618AQD (dual 12-bit DAC)	1

#	Component	Reference(s)	Specification	Qty
23	Microcontroller	A1	Arduino Nano Every	1
24	Rotary encoders w/ switch	— (off-board)	Mechanical quadrature encoder	2
25	TFT display module	— (off-board)	ILI9341 320×240 SPI + SD card slot	1

Table 2: Bill of Materials.

9. Result

The design has been validated stage-by-stage against the schematic and BOM: the voltage control loop's resistor-derived gain matches the firmware's calibrated gain within $\sim 4\%$, and the current-sense path's resistor-derived gain matches within $\sim 12\%$ (with the firmware's empirically measured value treated as authoritative).

Firmware for TFT bring-up, combined encoder/DAC/display operation, and the underlying SPI/DAC control-word derivation were each independently developed and are included in the project's code appendix.

10. Appendix

10.1 Code Files

- `adj_power_combined.ino` — full combined firmware (encoders, DAC, TFT, software current-limit foldback)
- `tft_bringup.ino` — standalone TFT bring-up / display test sketch

10.2 References

- TLC5618A dual 12-bit DAC datasheet
- LM2575-family adjustable buck regulator datasheet (output voltage design equation)
- Arduino Nano Every reference documentation
- ILI9341 TFT controller datasheet